Mismatched Pairs Dynamic Correction for Cross-Modal Alignment in Video Moment Retrieval

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***Abstract*—Video Moment Retrieval (VMR) aims to locate and identify specific moments in a video based on a natural language query. However, most existing methods overlook the issue of mismatched pairs (i.e., false negatives), wherein semantically similar or nearly identical queries and videos are erroneously treated as negative instances. To address this problem, we propose Mismatched Pairs Dynamic Correction (MPDC), a novel approach that rectifies such mismatched pairs on-the-fly without requiring additional annotations or intricate procedures. Specifically, we first construct both intra- and inter-modal similarity matrices for text-text, video-video, and text-video pairs, then aggregate these matrices to effectively detect previously overlooked mismatched pairs. We subsequently address these mismatches by adaptively updating their matching relationships during training, ensuring a dynamic and accurate correction process. Furthermore, to enhance the distinguishability of multi-modal features, we incorporate auxiliary queries derived from these similarity matrices to generate discriminative saliency scores, which serve as weights to refine the original feature representations. Our method is plug-and-play, seamlessly integrating into existing architectures and significantly boosting their performance. Extensive experiments conducted on four benchmark datasets (QVHighlights, Charades-STA, ActivityNet-Captions, and TACoS) demonstrate that our method achieves state-of-the-art performance.**

***Index Terms*—Video moment retrieval, mismatched pairs, cross-modal alignment.**

I. Introduction

VIDEO Moment Retrieval (VMR) is the task of locating and identifying specific moments in a video based on a given natural language query [1]. This task is inherently challenging, as it requires precise alignment between the query and the video content. Achieving such alignments is particularly difficult due to the complex nature of vision [2], [3], [4] and the diversity of user queries [5].

Recently, numerous methods have emerged to tackle the VMR task, each offering distinct approaches to align video and text features. One prominent approach is contrastive learning [6], [7], [8], where video and text features are aligned by bringing matching pairs closer in feature space while pushing non-matching pairs apart. Additionally, transformer-based models [9], [10] have shown great promise in improving the VMR task. These models typically perform cross-modal interaction to evaluate the relevance between video clips and textual queries, facilitating more accurate text-video matching.

However, most of the aforementioned methods largely overlook a critical issue: due to the inherent ambiguity in annotating semantic relevance and the subjective nature of manual labeling, many VMR datasets frequently contain semantically similar or nearly identical queries or videos incorrectly labeled as negative samples, which results in the creation of mismatched pairs [11], [12]. Because removing these mismatched pairs requires extensive manual inspection or sophisticated data cleaning, ensuring a completely noise-free dataset is rarely feasible. To enable the model to learn precise alignments between video and text, addressing this mismatch problem is a critical prerequisite. Nevertheless, most existing methods generally assume that each annotated query–video pair in the dataset constitutes a unique match [7], [10], [13], [14], overlooking the possibility of multiple queries or videos sharing highly similar content. As Fig. 1(a) illustrates, such overlooked instances are mistakenly treated as false negatives (i.e., mismatched pairs), significantly hindering model performance. In fact, as Fig. 2 shows, it is relatively common to encounter multiple semantically similar instances in current VMR datasets. Hence, ensuring the accuracy of training data is essential for achieving precise correspondences between queries and videos.

To this end, we propose a **M**ismatched **P**airs **D**ynamic **C**orrection (MPDC) framework that dynamically mitigates inherent mismatches in existing VMR datasets during training (Fig. 1(b)), enabling more accurate data utilization without additional annotations or complex procedures. MPDC leverages three complementary similarity matrices to relax rigid one-to-one video-text correspondences into flexible one-to-many associations. Text similarity benefits from recent advances in NLP, while video similarity remains challenging due to spatio-temporal complexity—making simple vector compression insufficient. To address this, we propose an event-driven similarity metric that segments videos temporally and aggregates inter-event similarities for robust comparison. For cross-modal alignment, text-video similarity is computed

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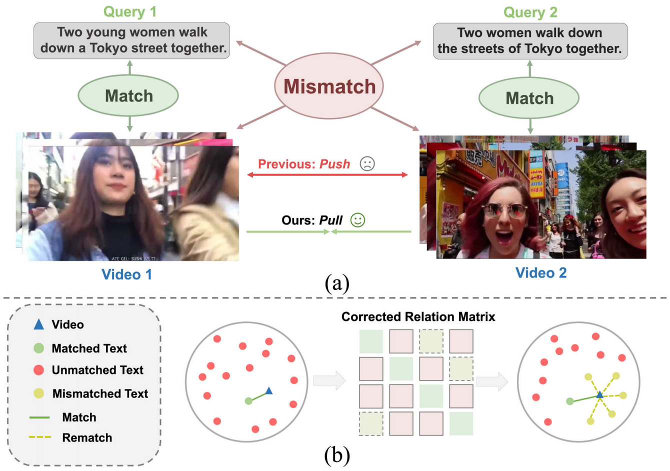


Fig. 1. (a) In existing VMR datasets, only annotated pairs are considered as matched, whereas numerous semantically similar but unannotated pairs are incorrectly regarded as mismatched (i.e., false negatives) and consequently pushed away during training. To address this issue, we propose a dynamic correction strategy to identify and mitigate such false negatives. (b) We construct a Corrected Relation Matrix that effectively adjusts training signals by alleviating the adverse effects of false negatives.

via learned multi-modal representations through cross-modal interactions in the encoder.

To enhance moment prediction by improving multi-modal feature discriminability, we leverage similarity matrices to identify auxiliary queries that complement the annotated query. These auxiliary queries, together with the video, are processed to generate distinctive saliency scores indicating relevance. Aggregating these scores as feature weights boosts feature distinction while suppressing noise. Thanks to its plug-and-play design, our framework integrates seamlessly into existing architectures, offering significant performance gains without structural changes.

Overall, we make the following contributions:

- We introduce MPDC, a dynamic correction framework that addresses mismatched pairs in the dataset during training by leveraging intra- and inter-modal similarity matrices. This plug-and-play framework can accurately identify mismatched pairs in the dataset.
- Based on the derived similarity matrices, we propose a method that leverages auxiliary queries to refine the original features with precise semantic relationships, thereby amplifying the distinguishability of multi-modal features.
- We conduct extensive experiments on four widely used benchmark datasets: QVHighlights, Charades-STA, ActivityNet-Captions and TACoS. The results demonstrate that our proposed method achieves state-of-the-art performance.

II. Related Work

A. Learning With Ambiguous Supervision

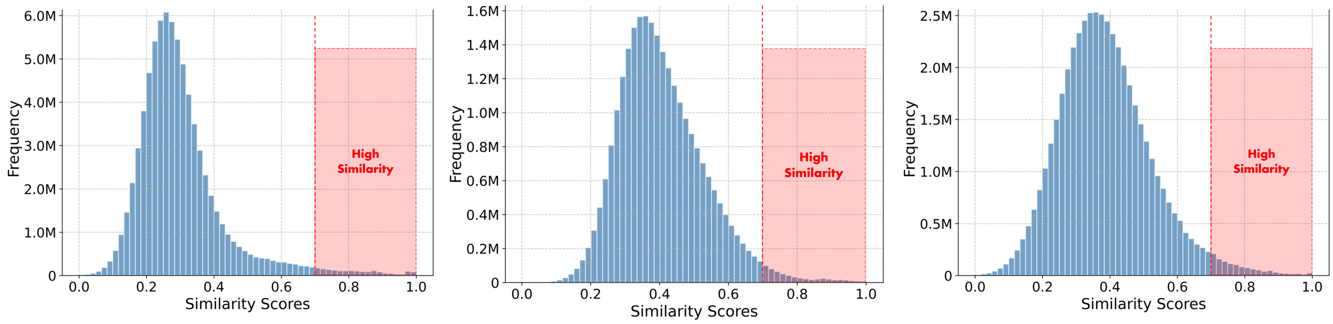
Learning robust representations under noisy or ambiguous supervision has been extensively studied in instance-level matching tasks such as image–text retrieval. Representative directions include hard or progressive negative mining [16], debiased or reweighted contrastive objectives that alleviate the influence of ambiguous negatives [17], [18], [19], and soft supervision strategies that replace binary labels with similarity-aware targets [20]. Related ideas have also been explored in weakly or sparsely supervised settings [21]. These approaches generally improve robustness by relaxing strict instance-level constraints or down-weighting uncertain negatives. However, directly applying these techniques to Video Moment Retrieval (VMR) is non-trivial. Unlike static instance matching, VMR requires fine-grained temporal grounding in untrimmed videos, where semantically similar moments may occur at different timestamps or correspond to different narrative roles. Consequently, supervision noise in VMR is not only instance-level but also temporally structured, making false negatives more pervasive and harder to identify.

B. Video Moment Retrieval

Video Moment Retrieval (VMR) aims to localize temporal segments in untrimmed videos that semantically align with a natural language query [9], [10], [22]. This task is inherently challenging, as it requires fine-grained cross-modal alignment between continuous video streams and free-form language descriptions, often under large variations in visual appearance, motion patterns, and linguistic expressions. Early VMR methods mainly follow a proposal-based paradigm [1], [23], [24], where candidate temporal segments are first generated using sliding windows or predefined anchors and then ranked according to query relevance. To avoid the limitations of handcrafted proposals, proposal-free approaches [25], [26] directly regress temporal boundaries in an end-to-end manner, improving efficiency and flexibility. More recently, transformer-based frameworks [9], [10], [27], [28], [29] have become dominant by modeling fine-grained video–text interactions through cross-modal attention and joint representations. These advances not only improve temporal localization accuracy but also benefit closely related tasks such as Highlight Detection [30]. Meanwhile, conditional video diffusion has been explored for fine-grained temporal grounding via boundary refinement [31], [32]. In parallel, low-annotation grounding has also been explored, e.g., hypotheses-tree based one-shot localization [33] and memory-guided few-shot semantic learning [34].

Despite significant architectural progress, most existing methods are still trained under a rigid one-to-one supervision assumption: within a mini-batch, each video–query pair is treated as the only positive match, while all other pairs are regarded as negatives. Such a design is convenient for contrastive or matching-based learning, but it implicitly assumes that annotations are complete and unambiguous. In practice, large-scale VMR datasets often contain semantically similar queries or video moments describing related events, leading to multiple potential matches beyond the annotated pair. Treating all unannotated pairs as negatives therefore introduces systematic false-negative noise, which may distort representation learning and hinder precise cross-modal alignment.

Recognizing the limitations of vanilla supervision, several recent works attempt to better utilize training signals in VMR. For example, [35] leverages multiple video–query pairs within



(a) Charades-STA (b) QVHighlights (c) TACoS

Fig. 2. We visualize the pairwise query similarity in above three datasets using Sentence Transformer [15]. All three datasets exhibit a substantial number of highly similar query pairs (in red region), which correspond to mismatched pairs (i.e., false negatives). Treating these as negatives, as in prior work, can substantially degrade model performance.

a mini-batch to model shared semantics across pairs. While such cross-pair modeling enhances representation learning, it does not explicitly identify potential positives under one-to-one supervision. Reference [36] constructs auxiliary positive and negative samples, often relying on large language models for data generation. Although these methods enrich training signals, they typically depend on additional data generation pipelines and do not directly address false negatives caused by incomplete temporal annotations. Beyond these, some studies improve robustness by alleviating dataset-level biases, e.g., [37] reduce the "0s bias" by introducing a circular competence-based captioner to re-balance temporally skewed supervision, while [38] enhance robustness with calibrated contrastive learning that down-weights faulty negatives and up-weights hard negatives; alternatively, [39] mitigates such biases via reweighting and sampling adjustment. However, these approaches do not explicitly model batch-wise mismatched text–video correspondences under rigid one-to-one annotations, which is the focus of our MPDC framework.

In contrast, we explicitly investigate *batch-wise mismatched pairs* in VMR training, referring to semantically consistent video–query pairs that are mistakenly treated as negatives under rigid one-to-one annotations. Our MPDC framework dynamically identifies such mismatched pairs and converts rigid supervision into flexible one-to-many associations on-the-fly, without requiring additional annotations. To the best of our knowledge, this is the first work that systematically addresses batch-wise false negatives arising from incomplete temporal annotations in VMR.

III. Method

A. Preliminary

To formalize the Video Moment Retrieval (VMR) and Highlight Detection (HD) tasks, given a video V and a text query Q, we represent the video as a sequence of clip features $F_v = \{v_i\}_{i=1}^{L_v} \in \mathbb{R}^{L_v \times D_v}$, where L_v denotes the number of video clips and D_v the feature dimension, and the query as a sequence of word features $F_t = \{q_i\}_{i=1}^{L_t} \in \mathbb{R}^{L_t \times D_t}$, where L_t is the query length and D_t the word feature dimension. For VMR, the objective is to localize the temporal segment relevant to Q, parameterized by its center m_c and duration m_σ. For HD, the model predicts clip-level saliency scores $S = \{s_i\}_{i=1}^{L_v}$, where each $s_i \in [0, 1]$ measures the relevance between the i-th clip and the query Q.

Baseline Framework: We adopt CG-DETR [27] as our baseline, a DETR-style framework for video temporal grounding. It aligns clip-level video and token-level query features through cross-modal interaction to localize the target moment. Its Adaptive Cross-Attention (ACA) module uses query-conditioned dummy tokens to divert attention from irrelevant clips, yielding more query-focused representations. Clip–word correlations are further injected into attention maps for fine-grained alignment. The fused features are then processed by a transformer encoder–decoder to predict temporal boundaries and saliency scores end-to-end.

B. Overview

To address mismatched pairs in video-text datasets, enhance cross-modal alignment precision, and boost feature discriminability via auxiliary texts, our Mismatched Pairs Dynamic Correction (MPDC) framework follows a coherent end-to-end workflow (see Fig. 3): paired videos and text queries are first encoded by dedicated visual/textual encoders, then fed into the Multi-modal Encoder (section III-C) for deep cross-modal interaction and joint representation learning; next, a Corrected Relation Matrix (built by fusing intra-modal (text-text, video-video) and inter-modal (text-video) similarities, section III-E) enables the Dynamic Correction of Mismatched Pairs module (section III-D) to identify false negatives and select auxiliary positive/negative texts; the Saliency-Aware Feature Amplifier (section III-F) then uses these auxiliaries to enhance query-relevant semantics and suppress noise; finally, corrected contrastive learning (section III-G) optimizes video-text alignment via the refined matrix, and the decoder localizes target moments—with the framework remaining plug-and-play, integrating seamlessly into existing architectures without extra annotations or complex preprocessing.

C. Multi-Modal Encoder

To align video and query features, we first project them into a shared d-dimensional space using separate MLPs. We then adopt the Adaptive Cross-Attention (ACA) module

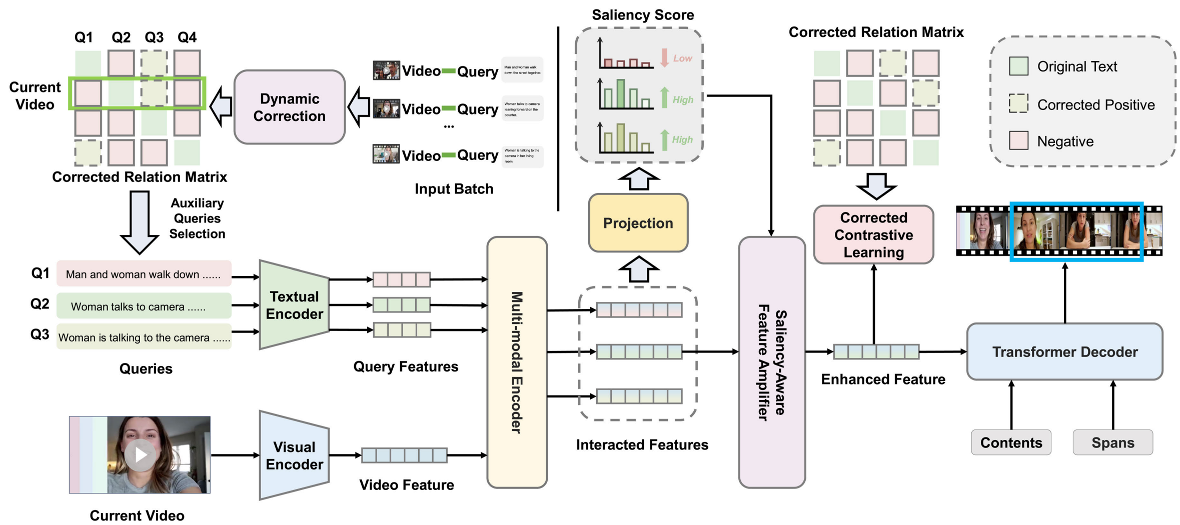


Fig. 3. Overview of the proposed MPDC. Based on the input video and text query, Dynamic Correction refines the matching relationship by assessing their semantic relations, while Auxiliary Query Selection identifies auxiliary texts with positive and negative samples to more comprehensively capture video semantics. The Multi-modal Encoder enables cross-modal interaction across all texts, and the Saliency-Aware Feature Amplifier uses these auxiliary features to enhance query-relevant semantics in the video features. Dynamic Correction is also applied for corrected contrastive learning to improve semantic accuracy. Finally, the Decoder outputs the predicted moment.

from [27], which highlights that not all video clips are equally relevant to the query semantics. Specifically, ACA introduces a set of learnable dummy tokens $D = [D_1, \dots, D_{L_d}]$ that are concatenated with query tokens in the attention key set, allowing dummy tokens to absorb attention weights from query-irrelevant clips and thus preventing irrelevant video content from being over-conditioned by the text query. Formally, ACA constructs

$$K = W_K([F_t; D]), \quad Q = W_Q(F_v), \quad V = W_V(F_t), \tag{1}$$

where W_* are learnable projection layers. We then compute fused video features via cross-attention while normalizing over both text tokens and dummy tokens:

$$F'_v = ACA(v_i) = \sum_{j=1}^{L_t} W_{i,j} V_j, \tag{2}$$

$$W_{i,j} = \frac{exp\left(\frac{Q_i K_j}{\sqrt{d}}\right)}{\sum_{k=1}^{L_t+L_d} exp\left(\frac{Q_i K_j}{\sqrt{d}}\right)}. \tag{3}$$

The fused features F'_v are further concatenated with the original query features F_t and passed through a stack of Transformer Encoder layers to enabledeeper cross-modal interaction. The final output, denoted as $\widetilde{F} = \{\widetilde{F}_v, \widetilde{F}_t\} \in \mathbb{R}^{(L_v+L_t)\times d}$, encapsulates the contextually enriched video features $\widetilde{F}_v$ and text features $\widetilde{F}_t$.

D. Dynamic Correction of Mismatched Pairs

In cross-modal text-video matching tasks, the construction of reliable positive/negative sample pairs is critical for guiding the model to learn discriminative semantic representations. Let $\mathcal{D} = \{\mathcal{V}, \mathcal{T}, \mathcal{M}\}$ denote a mini-batch of text-video samples, where $\mathcal{T} = \{Q_i\}_{i=1}^N$ and $\mathcal{V} = \{V_i\}_{i=1}^N$ represent the text and video subsets, respectively, with N being the batch size. The relation matrix $\mathcal{M} = \{m_{ij} | i, j = 1, \dots, N\} \in \mathbb{R}^{N\times N}$ encodes the pairwise matching labels: $m_{ij} = 1$ if the pair (V_i, Q_j) is semantically consistent (positive pair), and $m_{ij} = -1$ otherwise (negative pair). Most existing methods adopt a strict "one-to-one" annotation assumption, i.e., only diagonal elements $(i = j)$ in $\mathcal{M}$ are treated as positive pairs, while all off-diagonal elements $(i \neq j)$ are regarded as negative pairs. However, this simplification ignores the inherent semantic similarity among samples in real-world datasets: for instance, a video depicting "a dog chasing a ball" may share high semantic relevance with a text describing "a canine running after a toy", even if they are not annotated as a pair. Such off-diagonal pairs with actual semantic consistency are defined as "mismatched pairs" in this work (see Fig. 1(a)), and their misclassification as negative samples will introduce noisy supervision signals, thereby limiting the model's generalization ability.

To address this issue, we propose a dynamic correction strategy that adaptively rectifies mismatched pairs during the training process, aiming to construct a more accurate supervision signal. Specifically, we first define $S(\cdot, \cdot)$ as a similarity evaluator, which calculates the semantic similarity between samples based on the feature representations learned by the cross-modal encoder. In a given mini-batch, we further construct three core similarity matrices (all $\in \mathbb{R}^{N\times N}$):

- Video-video similarity matrix $S_{vv} = S(\mathcal{V}, \mathcal{V})$, where $S_{vv}[i, j]$ reflects the intra-modal semantic similarity between video V_i and V_j.
- Text-text similarity matrix $S_{tt} = S(\mathcal{T}, \mathcal{T})$, where $S_{tt}[i, j]$ quantifies the intra-modal semantic consistency between text Q_i and Q_j.
- Cross-modal text-video similarity matrix $S_{tv} = S(\mathcal{T}, \mathcal{V})$, where $S_{tv}[i, j]$ measures the inter-modal semantic alignment between text Q_i and video V_j.

The detailed construction of the three similarity matrices, including the feature normalization strategy and similarity

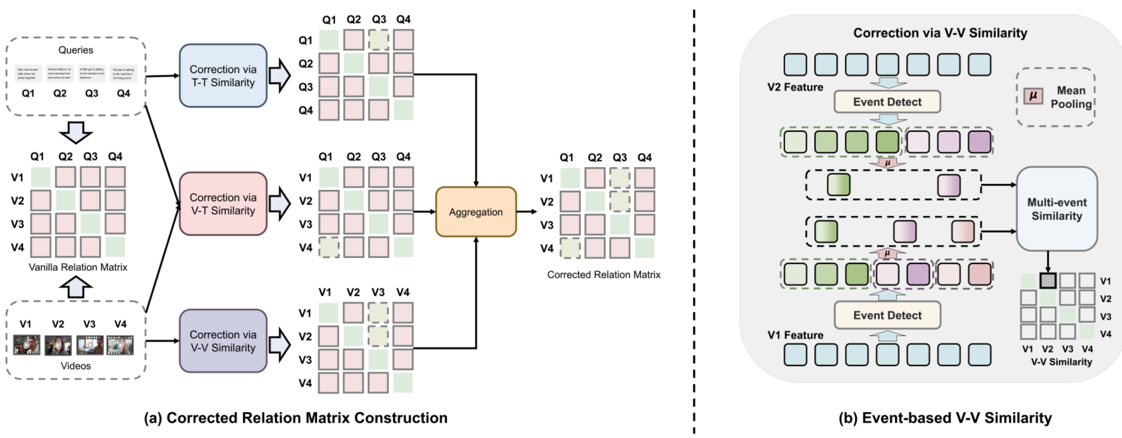


Fig. 4. (a) Illustration of the corrected relation matrix construction, including intra- and inter-modal similarity matrices for text-text (T-T), video-video (V-V), and video-text (V-T) pairs. (b) Illustration of the calculation for Event-based video-to-video (V-V) Similarity, where videos are segmented into events and similarity is computed based on inter-event relations.

metrics, is described in section III-E. By exploiting the complementary information from intra-modal and inter-modal similarities, we dynamically construct the corrected relation matrix $\mathcal{M}$ as:

$$\mathcal{M} = (S_{tv} > p_{tv}) \vee (S_{vv} > p_{vv}) \vee (S_{tt} > p_{tt}), \tag{4}$$

where $\vee$ denotes the logical *OR* operation, and p_{tv}, p_{vv}, p_{tt} are adaptive thresholds for detecting mismatched pairs. The threshold selection strategy, which includes data distribution–aware adjustment and validation-based optimization, is detailed in section IV-D.3.

In practice, the criterion $S_{tv} > p_{tv}$ is disabled in the early stage of training because insufficiently trained encoders tend to produce poorly aligned cross-modal features, which can introduce false positives. After convergence, this criterion is activated to further refine the correction results using more discriminative cross-modal representations.

In essence, the dynamic correction process supplements the original annotation labels with multi-modal similarity constraints, thereby redefining the matching relation m_{ij}: mismatched pairs originally labeled as -1 are corrected to 1, and each text or video sample is assigned multiple positive counterparts instead of a single annotated pair. This not only eliminates the noise caused by mislabeled negative samples but also enriches the positive sample space, providing more comprehensive supervision for the model. Fig. 4 (a) intuitively illustrates the transformation process from the original relation matrix to the corrected one.

Auxiliary Queries Selection: To enhance feature discriminability, we further select auxiliary positive and negative text queries for each video. For auxiliary positive query selection, one text Q_{pos} is randomly sampled from the positive set $\{Q_j \mid \mathcal{M}[i, j] = 1\}$ for each video V_i. For auxiliary negative query selection, we first construct an inverse similarity constraint matrix $\overline{\mathcal{M}}$ by reversing the threshold comparison in $\mathcal{M}$:

$$\overline{\mathcal{M}} = (S_{tv} < \overline{p}_{tv}) \vee (S_{vv} < \overline{p}_{vv}) \vee (S_{tt} < \overline{p}_{tt}), \tag{5}$$

where $\overline{p}_*$ are negative thresholds significantly lower than the corresponding p_* in $\mathcal{M}$. Based on $\overline{\mathcal{M}}$, we then similarly sample one text Q_{neg} from $\{Q_j | \overline{\mathcal{M}}[i, j] = 1\}$. Empirically, the negative threshold $\overline{p}_*$ is robust across datasets. In all experiments, we set $\overline{p}_* = 0.3$ to guarantee sufficient semantic dissimilarity for sampling Q_{neg}, and the performance is insensitive to its exact value within a reasonable range.

E. Similarity Matrices Construction

The similarity matrices S_{tt}, S_{vv} and S_{tv} are used in Dynamic Correction of Mismatched Pairs to accurately distinguish truly matched pairs.

1) Video-to-Video Similarity: A prevalent strategy for video-to-video similarity assessment involves compressing each video into a single vector, which often results in substantial information loss. To alleviate this limitation, as depicted in Fig. 4(b), we propose an event-based framework to capture similarity at a finer granularity. Inspired by [40] and [41], we leverage the Temporal Self-similarity Matrix (TSM) for event boundary estimation.

Specifically, we first construct the temporal self-similarity matrix $\mathbf{S} \in \mathbb{R}^{L_v \times L_v}$ by computing the cosine similarity between consecutive video representations F_v. Then, we define a contrastive kernel $\mathbf{Z} \in \mathbb{R}^{z \times z}$ with a kernel size of $z = 5$ as:

$$\mathbf{Z} = \begin{bmatrix} 1 & 1 & 0 & -1 & -1 \\ 1 & 1 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ -1 & -1 & 0 & 1 & 1 \\ -1 & -1 & 0 & 1 & 1 \end{bmatrix}. \tag{6}$$

This kernel is designed to emulate the boundary patterns inherent in the TSM. By convolving it with the diagonal elements of $\mathbf{S}$, we compute the boundary score vector $\mathbf{b} \in \mathbb{R}^{L_v}$. Using $\mathbf{b}$, we first discard scores below the mean boundary score $\overline{b}$, then apply a sliding max filter with a size of 3 to suppress consecutive redundant scores. The indices corresponding to the remaining non-zero scores are identified as event boundaries.

Subsequently, the event timestamps $\hat{P}$ are determined as the central coordinates and durations between consecutive boundary indices, based on which each video is segmented into events after thresholding. Specifically, guided by $\hat{P}$ and inter-frame similarity, the video is partitioned into discrete events, and each event's feature is obtained by mean-pooling the clip features within its corresponding time interval. We denote the event feature set of a video as $\boldsymbol{e} = e_1, e_2, \ldots, e_K \in \mathbb{R}^{K\times d}$, where K (the number of events) varies across different videos. The similarity between two videos V_a and V_b is then computed by averaging the pairwise cosine similarities between their respective event feature sets $\boldsymbol{e}_a$ and $\boldsymbol{e}_b$, which is formally defined as:

$$S_{vv}(V_a, V_b) = \text{Average}\left(\frac{\boldsymbol{e}_a \cdot \boldsymbol{e}_b^T}{\|\boldsymbol{e}_a\|_2 \|\boldsymbol{e}_b\|_2}\right). \tag{7}$$

2) Text-to-Text Similarity: Recent advancements in NLP allow for rapid and accurate text-to-text similarity measurement using pre-trained models. In our approach, we use SentenceBERT [15] to evaluate text similarity, defined as follows:

$$S_{tt}(Q_a, Q_b) = \text{SentenceBERT}(Q_a, Q_b). \tag{8}$$

3) Text-to-Video Similarity: Unlike the first two matrices that rely on pre-trained features, we compute the cosine similarity between the learned global video feature $\widetilde{F}_v^{global}$ and the text feature $\widetilde{F}_t^{global}$ (generated via the Saliency-Aware Feature Amplifier; see section III-F) to obtain the text-to-video similarity:

$$S_{tv}(T, V) = \frac{\widetilde{F}_t^{global} \cdot (\widetilde{F}_v^{global})^T}{\|\widetilde{F}_t^{global}\|_2 \|\widetilde{F}_v^{global}\|_2}. \tag{9}$$

F. Saliency-Aware Feature Amplifier

To further enhance the model's discriminative capability for query-relevant semantics, we design a saliency-aware feature amplifier that leverages auxiliary texts with corrected labels. For each video V, three associated texts are available: the original query Q_{ori}, the auxiliary positive text Q_{pos}, and the auxiliary negative text Q_{neg}. Each text query is paired with the same video and fed into the multi-modal encoder to obtain cross-modal interaction features:

$$\widetilde{F}_* = \{\widetilde{F}_v^*, \widetilde{F}_t^*\} = f_{enc}(V, Q_*), \quad * \in \{ori, pos, neg\}, \tag{10}$$

where f_{enc} denotes the multi-modal encoder. $\widetilde{F}_{ori}$ and $\widetilde{F}_{pos}$ correspond to interaction features between the video and its semantically relevant texts (Q_{ori} and Q_{pos}), while $\widetilde{F}_{neg}$ stems from the video's interaction with the irrelevant text Q_{neg}.

Subsequently, the video-specific interaction features $\widetilde{F}_v^{pos}, \widetilde{F}_v^{neg}$ and $\widetilde{F}_v^{ori}$ are input into an MLP layer to generate clip-wise saliency scores, which quantify the semantic relevance between each video clip and the corresponding text query:

$$\boldsymbol{s}_* = \{s_i^*\}_{i=1}^{L_v} = \text{MLP}(\widetilde{F}_v^*), * \in \{ori, pos, neg\}, \tag{11}$$

where $\boldsymbol{s}_*$ is the saliency score vector and s_i^* represents the saliency score of the i-th clip. To reinforce discriminative learning from auxiliary texts, we impose constraints: saliency scores of clips within the ground-truth moment should be high for $\boldsymbol{s}_{ori}$ and $\boldsymbol{s}_{pos}$, and low for $\boldsymbol{s}_{neg}$. Let $\overline{s}_{ori}$, $\overline{s}_{pos}$ and $\overline{s}_{neg}$ denote the average saliency scores of ground-truth moment clips in $\boldsymbol{s}_{ori}$, $\boldsymbol{s}_{pos}$ and $\boldsymbol{s}_{neg}$, respectively. The auxiliary loss is formulated as:

$$\mathcal{L}_{aux} = -\log\left(\frac{\overline{s}_{ori} + \overline{s}_{pos}}{2}\right) - \log(1 - \overline{s}_{neg}), \tag{12}$$

which refines saliency scores by leveraging clip-query relevance differences.

Given the high semantic similarity between Q_{ori} and Q_{pos}, their corresponding saliency scores $\boldsymbol{s}_{ori}$ and $\boldsymbol{s}_{pos}$ are integrated to enhance reliable relevance signals:

$$\boldsymbol{O} = Softmax(\boldsymbol{s}_{ori} + \boldsymbol{s}_{pos}). \tag{13}$$

The integrated score $\boldsymbol{O}$ is then used as a weighting factor to amplify the original video-text interaction $\widetilde{F}_v^{ori}$:

$$\widetilde{F}_v^{enhanced} = (\boldsymbol{O} \otimes \widetilde{F}_v^{ori}) \in \mathbb{R}^{L_v\times d}, \tag{14}$$

where $\otimes$ is element-wise multiplication. Attention pooling is further applied to $\widetilde{F}_v^{enhanced}$ and $\widetilde{F}_t^{ori}$ to extract d-dimensional global features $\widetilde{F}_v^{global}$ (video) and $\widetilde{F}_t^{global}$ (text).

Finally, $\widetilde{F}_v^{enhanced}$ is fed into the decoder for target moment prediction, while $\widetilde{F}_v^{global}$ (video) and $\widetilde{F}_t^{global}$ serve as inputs for Text-to-Video Similarity calculation (section III-E) and corrected contrastive learning (section III-G).

G. Corrected Contrastive Learning Loss

Our dynamic correction method mentioned in section III-D can re-assign multiple positives to a video or text, rather than just the original one. Therefore, to model accurate semantic alignment, we adopt a sigmoid loss [42] for our contrastive learning based on the corrected relation matrix $\mathcal{M} = \{m_{ij}|i, j = 1, \ldots, N\}$ and the text-video similarity matrix $S_{tv} = \{\sigma_{ij}|i, j = 1, \ldots, N\}$. This loss does not require computing global normalization factors and effectively turns the learning problem into the standard binary classification, which can be calculated as:

$$\mathcal{L}_{ccl} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N}\log\frac{1}{1 + e^{m_{ij}(-\sigma_{ij}/\tau + b)}}, \tag{15}$$

where τ and b are both temperatures, and m_{ij} indicates if the sample is positive ($m_{ij} = 1$) or negative ($m_{ij} = -1$).

H. Decoder and Learning Objectives

1) Decoder: In VMR, the decoder is a critical component that translates cross-modal interactive features into precise moment predictions. Recent advances in transformer-based decoders (e.g., DAB-DETR [43]) have highlighted that integrating task-specific positional information into query initialization not only speeds up convergence by providing meaningful initial anchors but also enhances prediction precision by aligning queries with target semantics. Drawing on this insight and following the query design paradigm of QD-DETR [10], we abandon the bounding box coordinates (typically used for spatial object detection) and instead initialize decoder queries with (m_c, m_σ)— where m_c denotes the center coordinate of the target moment and m_σ represents the moment's duration. To refine these initial queries and align them with the video-text cross-modal features, the (m_c, m_σ) parameters are further revised layer-wisely in the decoder, enabling iterative optimization toward the ground-truth moment.

2) Mr and HD Loss: The MR loss is used to measure the discrepancy between the prediction and ground truth moments. It consists of an L1 loss, a generalized IoU loss [44] and a cross-entropy loss:

$$\mathcal{L}_{mr} = \lambda_{L1}\|m - \widehat{m}\| + \lambda_{gIoU}\mathcal{L}_{gIoU}(m, \widehat{m}) + \lambda_{ce}\mathcal{L}_{ce}, \tag{16}$$

where m and $\widehat{m}$ are ground-truth moment and its corresponding moment containing center coordinate m_c and duration m_σ predicted by the decoder, respectively. $\mathcal{L}_{ce}$ is used to classify the predicted moment $\widehat{m}$ as either foreground (within the ground-truth moment) or background (located outside the ground-truth moment). For HD loss, we employ the loss $\mathcal{L}_{aux}$ in combination with the saliency loss $\mathcal{L}_{sal}$ from [27], which originally optimizes highlight detection with ranking-based objectives (e.g., margin ranking loss and rank contrastive loss) and an additional entropy regularization, and further includes additional objectives to supervise the learned query-attention correspondence and distill the inferred clip-word correlation into the cross-attention maps. The final loss is the sum of all the above losses:

$$\mathcal{L}_{overall} = \lambda_{mr}\mathcal{L}_{mr} + \lambda_{aux}\mathcal{L}_{aux} + \lambda_{sal}\mathcal{L}_{sal} + \lambda_{ccl}\mathcal{L}_{ccl}, \tag{17}$$

where each λ_* represents the corresponding weight for each loss component.

I. Inference

Benefiting from the filtering mechanism of our Dynamic Correction process during training, our model learns from higher-quality data, leading to improved performance. Consequently, no additional complex computations are required during inference. We simply input a single video and its corresponding text into the Multi-modal Encoder, multiply the extracted interactive features by their corresponding saliency scores to obtain enhanced representations, and then feed these enhanced features into the decoder. This process enables us to generate results with minimal computational overhead.

IV. Experiments

A. Datasets and Evaluation

1) Datasets: We evaluate our method on four mainstream VMR benchmarks: Charades-STA, QVHighlights, ActivityNet-Captions, and TACoS. Charades-STA contains 9,848 videos with 16,128 annotations; following prior work, we use the 11,166/1,242/3,720 train/val/test split and report results on the official test set. QVHighlights consists of 10,148 long videos with 7,218/1,150/1,542 queries for train/val/test, where a single query may correspond to multiple moments. ActivityNet-Captions, built on ActivityNet v1.3, contains long untrimmed videos with diverse temporal spans; we follow the official split with 37,417/17,505/17,031 moment–query pairs for train/val/test. TACoS includes 127 long cooking videos with approximately 19K sentence–moment pairs, in which target moments typically occupy a small portion of the video.

For feature extraction, we follow standard settings for fair comparison. On Charades-STA, we conduct experiments using both VGG features [45] and SlowFast features [46] combined with CLIP [47] (SF+C). For QVHighlights and TACoS, we

TABLE I
COMPARISON OF METHODS ON QVHIGHLIGHTS TEST SPLIT

Methods	MR R1 @0.5	MR R1 @0.7	MR mAP @0.5	MR mAP @0.75	MR mAP Avg.	HD Very Good mAP	HD Very Good HIT@1
XML [48]	41.83	30.35	44.63	31.73	32.14	34.50	55.30
MD-DETR [9]	52.89	33.02	54.82	29.40	30.73	35.70	55.60
UMT [49]	56.20	41.20	53.40	37.00	36.10	38.20	60.00
MH-DETR [50]	60.05	42.48	60.75	38.13	38.38	38.22	60.51
UniVTG [51]	58.86	40.86	57.60	35.59	35.47	38.20	60.96
QD-DETR [10]	62.40	44.98	62.52	39.88	39.86	38.94	62.40
MESM [2]	62.78	45.20	62.64	41.45	40.68	-	-
UVCOM [52]	63.55	47.47	63.37	42.67	43.18	39.74	64.20
CG-DETR [27]	65.43	48.38	64.51	42.77	42.86	40.30	66.20
TR-DETR [28]	64.66	48.96	63.98	**43.73**	42.62	39.91	63.42
LMR [53]	64.40	47.21	64.65	43.16	42.56	-	-
W2W [5]	64.66	47.34	64.68	43.71	43.11	40.35	64.79
Ours	**66.61**	**49.10**	**65.69**	43.28	**43.96**	**40.64**	**66.49**

adopt the SF+C features. For ActivityNet-Captions, we use CLIP features only. Text queries are encoded using CLIP for all datasets.

2) Evaluation: We follow standard evaluation protocols used in prior work for fair comparison. On QVHighlights, we evaluate both Video Moment Retrieval (MR) and Highlight Detection (HD). For MR, we report Recall@1 and mAP at IoU thresholds of 0.5 and 0.7, as well as mAP averaged over IoU thresholds from 0.5 to 0.95. For HD, we use mAP and HIT@1, where a prediction is considered positive only if its saliency score reaches the *Very Good* level. On Charades-STA, ActivityNet-Captions, and TACoS, MR performance is evaluated using Recall@1 at IoU thresholds of 0.3, 0.5, and 0.7, together with mean IoU (mIoU).

B. Implementation Details

Following CG-DETR [27], we use the same loss weights with $\lambda_{L1} = 10$, $\lambda_{gIoU} = 1$, and $\lambda_{ce} = 4$. The weights λ_{ccl} and λ_{aux} are set to (1, 0.2) for QVHighlights, (1, 0.1) for Charades-STA and ActivityNet-Captions, and (0.5, 0.2) for TACoS, while λ_{sal} follows CG-DETR. We adopt large batch sizes to mine sufficient mismatched pairs, using 128 for QVHighlights, Charades-STA, and ActivityNet-Captions, and 64 for TACoS. The learning rate is set to 1e−4 for QVHighlights and 2e−4 for the other datasets. All experiments are conducted on a single NVIDIA A100 GPU.

C. Comparison With State-of-the-Arts

To comprehensively validate the effectiveness and generalization of our MPDC framework, we conduct extensive comparisons with state-of-the-art (SOTA) methods on four mainstream VMR benchmarks. Specifically, we evaluate the Video Moment Retrieval (MR) task on the *test* splits of Charades-STA and TACoS, as well as the *val_2* split of ActivityNet-Captions (results in tables II to IV), and the joint tasks of MR and Highlight Detection (HD) on the QVHighlights *test* split (results in table I).

1) Results on QVHighlights: As a benchmark focusing on joint MR-HD tasks, QVHighlights poses challenges of single queries mapping to multiple moments and diverse video

TABLE II
COMPARISON OF METHODS ON CHARADES-STA TEST SPLIT

Methods	Feature	R1	
		@0.5	@0.7
MMN [24]		47.31	27.28
G2L [8]		47.91	28.42
MD-DETR [9]		52.07	30.59
UniVTG [51]		52.89	33.02
QD-DETR [10]		55.51	34.17
TR-DETR [28]	VGG	53.57	30.81
UVCOM [52]		54.57	34.13
CG-DETR [27]		55.22	34.19
W2W [5]		54.49	33.68
LMR [53]		55.91	35.19
Ours		**57.01**	**36.23**
2D-TAN [7]		46.02	27.50
MD-DETR [9]		53.63	31.37
UniVTG [51]		58.01	35.65
QD-DETR [10]		57.31	32.55
TR-DETR [28]	SF+C	57.61	33.52
W2W [5]		58.33	34.68
CG-DETR [27]		58.44	36.34
LLMEPET [54]		-	36.49
UVCOM [52]		59.25	36.64
Ours		**59.39**	**37.47**

TABLE III
COMPARISON OF METHODS ON ACTIVITYNET CAPTIONS VAL_2 SPLIT

Methods	Feature	R1	
		@0.5	@0.7
MD-DETR [9]		36.10	20.40
QD-DETR [10]		36.90	21.40
UVCOM [52]	CLIP	37.00	21.50
Vid-Group [55]		37.83	18.57
CG-DETR [27]		38.80	22.60
TRACE (7B) [56]	-	37.70	**24.00**
Ours	CLIP	**39.97**	23.68

TABLE IV
COMPARISON OF METHODS ON TACOS TEST SPLIT

Methods	R1		
	@0.3	@0.5	@0.7
2D-TAN [7]	40.01	27.99	23.92
MD-DETR [9]	37.97	24.67	11.97
G2L [8]	42.74	30.95	11.97
UniVTG [51]	51.44	34.97	17.35
UVCOM [52]	-	36.39	23.32
LLMEPET [54]	52.73	-	22.78
MESM [2]	52.69	39.52	-
R^2-Tuning [57]	49.71	38.72	**25.12**
LMR [53]	53.24	37.16	22.81
CGDETR [27]	54.40	39.50	23.40
Ours	**54.78**	**40.01**	24.26

content. As shown in table I, our MPDC achieves consistent improvements across all metrics, outperforming all competing methods. For the MR task, our method raises R1@0.5 from 65.43 (CG-DETR [27]) to 66.61 (+1.2%), R1@0.7 from 48.38 to 49.10 (+0.7%), and the average mAP from 42.86 to 43.96 (+1.1%). For the HD task, the mAP reaches 40.64 and HIT@1 hits 66.49, achieving an average 1% gain over SOTA.

TABLE V
ABLATION STUDY OF COMPONENTS ON QVHIGHLIGHTS VAL SPLIT

Method	R1@0.5	R1@0.7	mAP
Base	63.04	48.34	41.02
+ CCL	65.29	49.98	43.47
+ CCL + SAFA	**68.45**	**52.78**	**45.82**

This performance confirms that MPDC effectively mitigates mismatched pairs and enhances cross-modal alignment, supporting robust performance on both tasks.

2) Results on Charades-STA: Charades-STA is widely used for evaluating MR accuracy, with tests under two common feature backbones (VGG and SF+C). As illustrated in table II, our MPDC maintains superior performance across both backbones. Under the VGG feature, at the strict IoU threshold R1@0.7, we achieve 36.23, outperforming QD-DETR [10] (34.17) by 2.06% and CG-DETR [27] (34.19) by 2.04%. Under the more powerful SF+C feature, our R1@0.5 (59.39) and R1@0.7 (37.47) exceed UVCOM [52] (59.25/36.64) and CG-DETR (58.44/36.34), demonstrating the framework's adaptability and effectiveness across different feature encoders. We note that the gains under SF+C are relatively modest due to a limited-headroom effect: strong 3D spatiotemporal features and cross-modal representations already yield competitive baselines, leaving limited-headroom for large absolute improvements. Meanwhile, MPDC targets an issue orthogonal to feature quality—dataset-level false negatives induced by incomplete annotations under contrastive learning—thereby providing cleaner supervision and enabling more effective cross-modal alignment.

3) Results on ActivityNet Captions: To further validate scalability, we conduct experiments on the large-scale ActivityNet Captions *val_2* split. As shown in table III, our method achieves 39.97 (R1@0.5) and 23.68 (R1@0.7), outperforming strong baselines under the same setting with CLIP features. Notably, our method also shows competitive performance against TRACE-7B, achieving higher R1@0.5 while remaining comparable at IoU=0.7. These results indicate that our method generalizes effectively to large-scale benchmarks.

4) Results on TACoS: TACoS is a challenging dataset with long videos (avg. 287s) and target moments occupying a tiny fraction of the total duration. As shown in table IV, our MPDC achieves competitive results among SOTA methods, with modest improvements over CG-DETR [27]. This limited gain is attributed to two factors: first, TACoS only contains 127 videos (far fewer than Charades-STA's 1,334 and QVHighlights' 10,148), and second, it has fewer highly similar text pairs (see Fig. 2), leading to insufficient mismatched pairs for the dynamic correction module to mine effective auxiliary samples.

D. Ablation Study

To assess the effectiveness of key components in our method, we perform a comprehensive ablation study.

1) Effect of Each Component: We analyze the contribution of each component of our method in table V. CCL refers

TABLE VI
ABLATION STUDY OF S_{tt}, S_{vv}, AND S_{tv} ON THE QVHIGHLIGHTS VAL SPLIT

S_{tt}	S_{vv}	S_{tv}	R1@0.5	R1@0.7	mAP
✓			67.51	51.98	44.86
✓	✓		67.93	51.74	45.01
✓	✓	✓	**68.45**	**52.78**	**45.82**

TABLE VII
EFFECT OF AUXILIARY TEXT SAMPLING QUANTITY ON PERFORMANCE AND TRAINING EFFICIENCY ON THE QVHIGHLIGHTS VAL SPLIT. k_{pos} AND k_{neg} DENOTE THE NUMBERS OF SAMPLED POSITIVE AND NEGATIVE TEXTS PER VIDEO. #ENC. INDICATES THE NUMBER OF MULTI-MODAL ENCODER FORWARD PASSES PER VIDEO, AND REL. TIME REPORTS THE RELATIVE TRAINING TIME NORMALIZED TO THE 0-1 SETTING

k_{pos}	k_{neg}	mAP	#Enc.	Rel. Time
0	1	44.89	2	1.00×
1	0	44.24	2	1.00×
1	1	<u>45.82</u>	3	1.46×
2	2	**46.07**	5	2.38×
3	3	45.68	7	3.30×

to the Corrected Contrastive Learning Loss built upon the corrected relation label $\mathcal{M}$, while SAFA denotes the Saliency-Aware Feature Amplifier. For fair comparison, the Base model adopts the standard single-query setting and uses only the annotated query Q_{ori}. As no correction is applied and $\mathcal{M}$ is not constructed, the model is optimized solely with the MR loss and the saliency loss derived from Q_{ori}. When CCL is introduced, Dynamic Correction is activated to construct $\mathcal{M}$ and enable corrected contrastive learning, which alleviates the false-negative issue. Building upon this, adding SAFA further incorporates auxiliary positive and negative queries selected from $\mathcal{M}$ to form a multi-query setting. This design refines video–text representations through saliency-aware amplification and auxiliary saliency supervision. As reported in table V, CCL alone brings improvements of 1.5–2.1%. Combining CCL with SAFA leads to larger gains, yielding a 5% increase on R1@0.5 and a 4.3% increase on R1@0.7.

2) Effect of Different Similarity Matrices: We investigate the roles of S_{tt}, S_{vv} and S_{tv} mentioned in section III-D. As table VI shows, when filtering with S_{tt} alone, our method already shows promising performance, and optimal performance is achieved when all three matrices are combined.

3) Effect of Auxiliary Text Sampling: We study the impact of auxiliary positive and negative texts on model performance, with results reported in table VII. Compared with using only the original query, introducing either auxiliary positive or negative texts consistently improves localization performance, demonstrating their complementary roles in enhancing feature discriminability. Notably, jointly using one positive and one negative auxiliary text per video (i.e., the 1–1 setting adopted in our final model) achieves a favorable balance between performance and efficiency. This indicates that a balanced incorporation of semantically similar and dissimilar texts provides more informative supervision, enabling the model to better emphasize query-relevant semantics while suppressing irrelevant content.

TABLE VIII
THE ABLATION STUDY OF DIFFERENT VIDEO-VIDEO SIMILARITY METHODS ON THE QVHIGHLIGHTS VAL SPLIT

video-video Similarity	R1@0.5	R1@0.7	mAP
Mean-Pooling	66.76	49.59	43.71
Max-Pooling	65.43	47.96	42.15
Uniform Sampling + Pairwise Similarity	68.03	51.45	45.01
Event-Based (Ours)	**68.45**	**52.78**	**45.82**

TABLE IX
THE STATISTICS OF SIMILARITY MATRICES S_{tv}, S_{vv} AND S_{tt} ON QVHIGHLIGHTS TRAIN SPLIT

Similarity Matrix	Mean (μ)	Std Dev (σ)	Initial Threshold ($\mu + 2\sigma$)
S_{tv}	0.37	0.24	p_{tv} = 0.85
S_{vv}	0.40	0.22	p_{vv} = 0.84
S_{tt}	0.36	0.21	p_{tt} = 0.78

4) Different Methods for Calculating V-V Similarity: Table VIII compares our event-based video-video similarity with several commonly used aggregation strategies, including Mean-Pooling, Max-Pooling, and Uniform Sampling + Pairwise Similarity. Mean- and Max-Pooling compress all clip features into a single global representation. For Uniform Sampling + Pairwise Similarity, we uniformly sample 16 frames from each video and compute the video-level similarity by averaging all pairwise cosine similarities between the sampled frame features. In contrast, our event-based method explicitly models inter-event relations and consistently achieves the best performance, demonstrating its effectiveness for V-V similarity estimation.

5) Thresholds p_* for Positive Sampling: Our dynamic correction does not rely on finely tuned thresholds; instead, it only requires them to lie in a reasonable high-confidence range. We adopt a two-step strategy to determine p_*. First, each threshold is initialized based on the statistics of the corresponding similarity matrix S_{tv}, S_{vv}, or S_{tt}: $p_* = \mu(S_*) + 2\sigma(S_*)$, where $\mu(\cdot)$ and $\sigma(\cdot)$ denote the mean and standard deviation. Statistics on QVHighlights are reported in table IX, and thresholds for other datasets are computed in the same manner. Second, we apply a lightweight grid search to refine the thresholds within a high-confidence range.

As shown in Fig. 5, the performance remains stable when thresholds fall within a broad interval (approximately 0.85–0.95) across all three benchmarks. Within this range, selected pairs are already sufficiently similar, so varying the thresholds only marginally affects the corrected positives and leads to negligible performance changes. In contrast, lower thresholds introduce noisy corrections, while overly high thresholds gradually reduce the correction effect and approach the original annotation setting. Therefore, the grid search mainly serves to select an optimal configuration within a stable region, and the overall framework is insensitive to dataset-specific threshold choices.

6) Thresholds $\overline{p}_*$ for Negative Sampling: For sampling Q_{neg}, we fix the negative threshold as $\overline{p}_* = 0.3$ across all datasets. An ablation study in table X shows that performance is insensitive to this choice. When $\overline{p}_*$ is sufficiently small, the

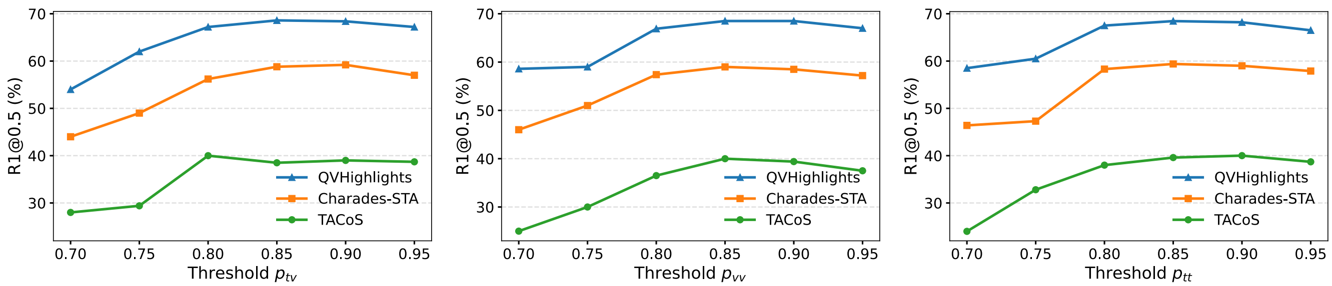


Fig. 5. Sensitivity analysis of thresholds p_{tv}, p_{vv} and p_{tt} in MPDC on three datasets. One threshold is varied while the other two are fixed. Performance remains stable in a high-confidence range (0.85–0.95), suggesting that our method is robust to the exact choice of thresholds.

TABLE X
THE ABLATION STUDY OF DIFFERENT VALUES ON $\overline{p}_*$ ON QVHIGHLIGHTS VAL SPLIT

$S_{tv} < \overline{p}_{tv}$	$S_{vv} < \overline{p}_{vv}$	$S_{tt} < \overline{p}_{tt}$	mAP
0.5	0.5	0.5	44.26
0.4	0.4	0.4	44.98
0.3	0.3	0.3	**45.82**
0.2	0.2	0.2	45.68
0.1	0.1	0.1	45.56

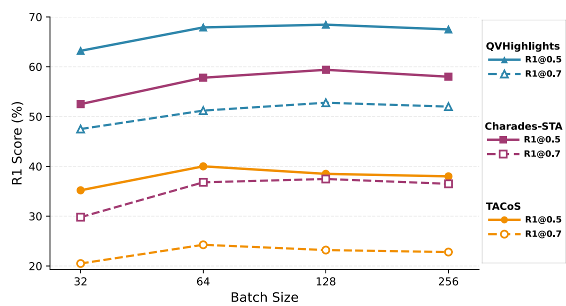


Fig. 6. Effect of batch size on retrieval performance (R1@0.5 and R1@0.7) under fixed thresholds. Increasing the batch size mainly improves the *recall* of potential positive samples, while performance saturates once sufficient candidates are included (128 for QVHighlights and Charades-STA, and 64 for TACoS).

sampled negative texts are already semantically dissimilar to the target video.

7) Effect of Batch Size: We ablate the batch size under fixed thresholds, as shown in Fig. 6. Increasing the batch size from 32 improves retrieval performance because it increases the recall of potential positive samples (i.e., more candidates are included for correction), but the gain saturates once sufficient candidates are covered, and further enlarging the batch yields only marginal improvement. Overall, these results suggest that batch size mainly affects *recall* (candidate coverage), whereas the thresholds p_* determine the *precision* of the screening process.

8) Plug-and-Play Capability: In table XI, we demonstrate the plug-and-play capability of MPDC by integrating it into two representative VMR frameworks, BM-DETR [58] and EaTR [41], and evaluating on the QVHighlights *val* split (results reproduced for BM-DETR are marked with ‡ since validation results are not reported). MPDC is attached on top of the original video–text representations of both frameworks without modifying their encoder–decoder localization pipelines. During training, a Corrected Relation Matrix M is constructed from text–text, video–video, and text–video similarities to identify false negatives, based on which auxiliary positive and negative queries are sampled, introducing only two additional objectives: the Corrected Contrastive Learning (CCL) loss and auxiliary-query saliency supervision. The video features are further refined via saliency-aware reweighting using $s^{ori} + s^{pos}$, which enhances query-relevant semantics while suppressing irrelevant patterns guided by the negative query. Specifically, for EaTR, MPDC is injected into the event-aware cross-modal features immediately before span decoding, allowing the corrected relations and auxiliary queries to reweight the grounding head while keeping its architecture unchanged, whereas for BM-DETR, MPDC is applied to the DETR-style video–text interaction features prior to moment prediction, enabling the decoder to attend over saliency-corrected, query-specific representations derived from both original and auxiliary saliency signals.

TABLE XI
PLUG-AND-PLAY CAPABILITY ON QVHIGHLIGHTS VAL SPLIT. ‡ DENOTES THE RESULTS REPRODUCED BY US

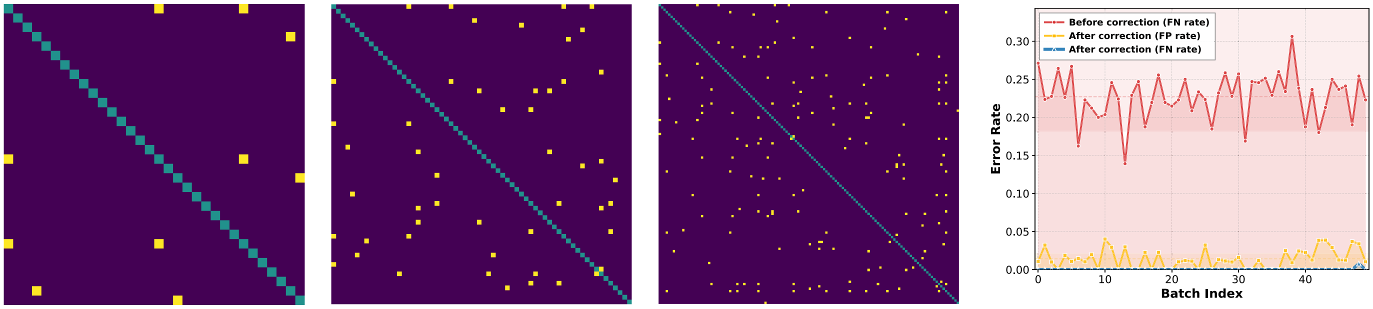
Method	R1@0.5	R1@0.7	mAP
EaTR	61.36	45.79	41.74
EaTR + Ours	**63.89**	**47.94**	**42.23**
BM-DETR‡	58.06	41.68	37.76
BM-DETR‡ + Ours	**61.29**	**44.58**	**39.91**

TABLE XII
OVERHEAD OF SIMILARITY MATRIX CONSTRUCTION UNDER BATCH SIZE $B = 128$. TIME IS MEASURED IN MS/ITER, AND COMPUTATIONAL COST IS REPORTED IN GFLOPS/ITER

Matrix	Time	GFLOPs
S_{tt}	0.033	0.008
S_{tv}	0.041	0.008
S_{vv}	24.109	0.422

E. Computational Complexity and Efficiency

We report the computational overhead of MPDC in terms of runtime (ms/iter) and GFLOPs (GFLOPs/iter) under the same experimental setting as the baseline CG-DETR. Table XII summarizes the cost of constructing the similarity matrices for



(a) $\mathcal{M}$, B=32 (b) $\mathcal{M}$, B=64 (c) $\mathcal{M}$, B=128 (d) FN/FP before vs. after correction

Fig. 7. **Left:** Visualization of corrected relation matrices $\mathcal{M}$ on Charades-STA with different batch sizes ($B \in \{32, 64, 128\}$). Diagonal pairs (green) are original annotations; corrected mismatches (yellow); unmatched pairs (dark blue). **Right:** False negative (FN) and false positive (FP) rates before and after applying the proposed dynamic correction on Charades-STA. Results are computed over 50 randomly sampled training batches (batch size = 128). After correction, the FN rate is substantially reduced, with only a small number of FPs introduced.

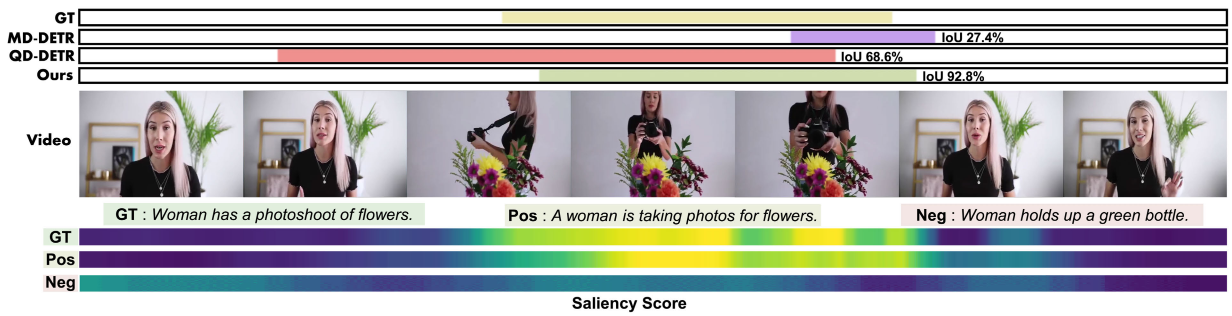


Fig. 8. Qualitative results of our method on the QVHighlights.

TABLE XIII

Efficiency Comparison. B Denotes the Batch Size. Time Is Measured in ms/Iter, and Computational Cost Is Reported in GFLOPs/Iter

	Training			Inference		
Method	B	Time	GFLOPs	B	Time	GFLOPs
CG-DETR	32	137.02	18.17	32	80.77	14.21
CG-DETR	128	171.40	72.42	32	80.77	14.21
Ours	128	363.61	132.10	32	80.85	14.21

building the corrected relation matrix $\mathcal{M}$. The results show that S_{tt} and S_{tv} introduce negligible overhead, while most additional cost comes from S_{vv} due to the event-based video-video similarity computation.

We further compare the overall efficiency in Table XIII. The additional training overhead mainly comes from the construction of the corrected relation matrix $\mathcal{M}$ and the auxiliary query selection for saliency-aware feature amplification. Although MPDC introduces extra training complexity, it consistently improves retrieval performance as shown in our main results, and it does not affect test-time deployment since the inference time and GFLOPs remain almost unchanged.

V. Qualitative Analysis

A. FN–FP Trade-off Analysis of Dynamic Correction

We quantify the FN–FP trade-off introduced by our dynamic correction, where FN denotes true matches incorrectly treated as negatives and FP denotes incorrect matches mistakenly treated as positives. On Charades-STA, we randomly sample 50 training batches (batch size = 128), covering nearly half of the training set, and construct the corrected relation matrix using the same procedure as in training. We then use GPT-4o to check whether the auxiliary queries selected by the corrected matrix are semantically consistent with the corresponding original GT query, and compute the resulting FN/FP ratios (Fig. 7(d)). Before correction, the average FN rate is 22.82%, while after correction the combined error rate (FP+FN) drops to approximately 0.93%, indicating that our method removes most harmful FNs with only negligible FPs introduced. We further report mismatch statistics on the evaluated pairs, where *mismatch* is defined as the total count of FP+FN. As summarized in table XIV, the mismatch rate drops from 22.82% (1915/8391) before correction to 0.93% (78/8391) after correction, corresponding to a 95.9% relative reduction. Notably, dynamic correction introduces only 77 false positives while almost completely eliminating false negatives (FN reduces to 1).

B. Visualization of Grounding

In Fig. 8, for the ground-truth sentence "Woman has a photoshoot of flowers", our MPDC method—enhanced by auxiliary texts—outperforms MD-DETR and QD-DETR in moment localization. QD-DETR fails to capture the core action (photoshoot), while MD-DETR cannot localize the complete ground-truth moment. In contrast, our approach

(a) Multi-event dependency. (b) Intent-level ambiguity.

Fig. 9. Visualization of two representative failure cases in the correction process. Left: multi-event dependency (runs out vs. runs back out). Right: intent-level ambiguity (begins sneezing vs. pretends to start sneezing). These cases share high lexical similarity but differ in subtle semantics, which may lead to occasional over-corrections.

TABLE XIV

Mismatch Statistics Before and After Dynamic Correction on Charades-STA. "Pairs" Is the Number of Evaluated Sample Pairs (50 Randomly Sampled Training Batches, Batch Size = 128); "FP"/"FN" Are the False-Positive/False-Negative Counts, "Mismatch" = FP+FN, and "Rate" Is the Mismatch Ratio. "FP" Is "–" Before Correction Since Only the Original Pairs Are Considered

Stage	Pairs	FP	FN	Mismatch	Rate↓
Before	8,391	–	1,915	1,915	22.82%
After	8,391	77	1	78	0.93%

leverages auxiliary text-based discriminative learning to produce more coherent predictions and accurately pinpoint target segments. The relevance scores from original/positive/negative texts further reflect video-query alignment, enabling fine-grained distinction between similar frames.

C. Visualization of $\mathcal{M}$ With Different Batch Sizes

As Fig. 7(a-c) shows, we visualize the corrected relation matrix $\mathcal{M}$ (obtained in section III-D) of the Charades-STA dataset under the setting of $p_{tv} = 0.90, p_{vv} = 0.85, p_{tt} = 0.85$ and different batch sizes (32, 64 and 128). In the visual representations, the original annotated pairs from dataset (elements on the diagonal) are denoted in *green* and corrected mismatched pairs by our method are denoted in *yellow*, while unmatched pairs are shown in *dark blue*. It can be observed that when the batch size is set to 32, only a limited number of mismatched pairs are identified for each sample. As the batch size increases, the number of mismatched pairs associated with each sample gradually increases. Particularly, when the batch size is set to 128, a significant number of mismatched pairs can be observed within the batch, which would severely hinder model learning if vanilla contrastive learning methods were applied.

D. Visualization of Failure Case

Although MPDC effectively corrects many false negatives, it may occasionally introduce over-corrections when two queries are lexically similar but semantically different. As shown in Fig. 9(a), "person runs out" and "person runs back out" share similar exit motions, but "runs back out" implies a multi-event history (running in first and then exiting again), whereas "runs out" denotes a direct exit. Similarly, in Fig. 9(b), "person begins sneezing" and "a person pretends to start sneezing" are nearly identical in wording, yet differ in subtle intent-level cues that are hard to capture from clip-level features. These cases suggest that similarity-driven correction is less reliable for intent- or history-sensitive queries, and could be improved with stricter semantic constraints or event-history modeling.

VI. Conclusion

The paper introduces MPDC, a novel framework that rectifies mismatched text-video pairs within datasets during the training phase. By constructing inter- and intra-modal similarities for text-text, video-video, and text-video pairs, our method effectively addresses the issue of mismatched pairs, enabling more accurate alignment between video and text representations without the need for additional annotations. Furthermore, to improve feature distinguishability, we introduce auxiliary queries that refine the original features through saliency-aware weighting, improving the model's focus on relevant information. MPDC is particularly effective under dense, semantically overlapping queries/moments with incomplete one-to-one annotations, where contrastive learning suffers from frequent false negatives; it corrects them on-the-fly to provide cleaner and more stable supervision, while remaining plug-and-play and keeping inference almost unchanged. Extensive experimental evaluations demonstrate that our approach significantly boosts performance, setting a new standard for video moment retrieval.

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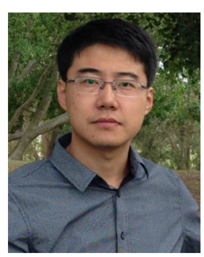

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